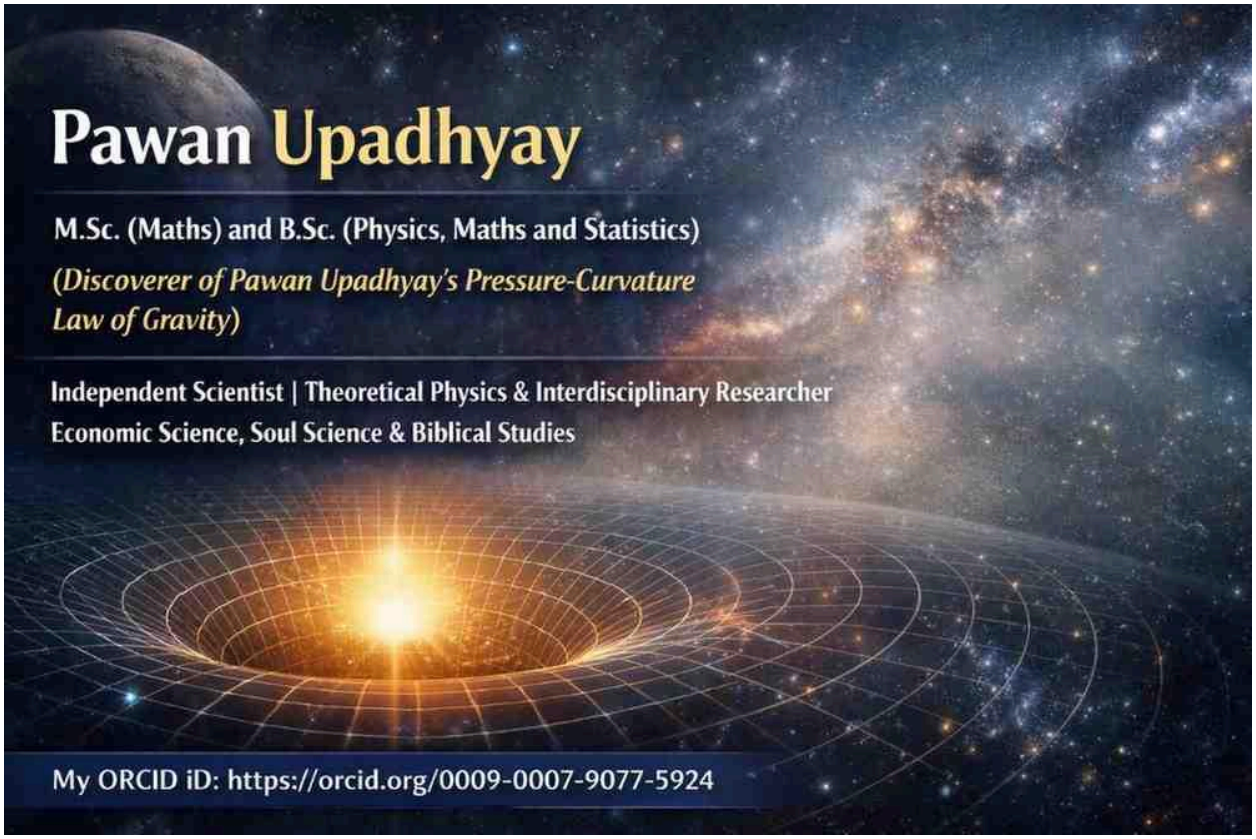
THE LORENTZ-INVARIANT CLASSICAL WAVE EQUATION FOR A MASSLESS SCALAR FIELD

A Four-Dimensional Formulation in Minkowski Spacetime

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ABSTRACT

The classical wave equation is traditionally written as

$\partial^2\Phi/\partial t^2 = c^2 \nabla^2 \Phi$

and is commonly associated with the propagation of disturbances through a physical medium.

However, the mathematical structure of the wave equation is more general than mechanical wave motion. When formulated in four-dimensional spacetime using the coordinates (x, y, z, ct), the equation takes the Lorentz-invariant form

$$\Box\Phi = 0$$

This equation describes the propagation of a free massless scalar field in Minkowski spacetime.

This paper examines the mathematical transition from the conventional three-dimensional wave equation to its four-dimensional relativistic form. The spacetime metric, d'Alembertian operator, plane-wave solutions, dispersion relation, null propagation, and massless character of the field are developed. The distinction between mechanical waves and classical field waves is also examined. Finally, the scalar wave equation is compared with the electromagnetic wave equations derived from Maxwell's theory.

1. INTRODUCTION

The wave equation occupies a central position in classical mathematical physics. It appears in the theory of vibrating strings, acoustics, elasticity, electromagnetism, and many other areas of physics.

In its familiar form, the scalar wave equation is

$$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$$

where Φ represents the quantity or field undergoing wave propagation and c represents the characteristic propagation velocity.

In mechanical systems, Φ may represent a displacement, pressure variation, or another physical quantity associated with matter. Such examples naturally lead to the idea that a wave is a disturbance of a material medium.

The development of relativistic physics provides a broader perspective. A field can possess wave dynamics without representing the mechanical displacement of a material substance. In particular, the electromagnetic field propagates through vacuum and is described classically by Maxwell's equations.

The purpose of this paper is to investigate the four-dimensional formulation of the classical wave equation using the spacetime coordinates

(x, y, z, ct).

The central equation considered is

$$\Box\Phi = 0$$

where $\Box$ denotes the d'Alembertian operator.

The objective is not to replace the ordinary wave equation but to demonstrate that its relativistic structure is naturally expressed through a four-dimensional spacetime operator.

2. THE CLASSICAL WAVE EQUATION

Let

$\Phi = \Phi(x, y, z, t)$

be a scalar field.

The three-dimensional Laplacian is

$\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$

The classical wave equation is therefore

$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$

Expanding the Laplacian,

$\partial^2\Phi/\partial t^2 = c^2(\partial^2\Phi/\partial x^2 + \partial^2\Phi/\partial y^2 + \partial^2\Phi/\partial z^2)$

Dividing by c^2 and rearranging,

$\partial^2\Phi/\partial x^2 + \partial^2\Phi/\partial y^2 + \partial^2\Phi/\partial z^2 - (1/c^2)\partial^2\Phi/\partial t^2 = 0$

This equation already contains the mathematical relationship between spatial variation and temporal variation that characterizes wave propagation.

3. FOUR-DIMENSIONAL SPACETIME COORDINATES

We introduce the four coordinates

$x^\mu = (x, y, z, ct)$

The field becomes

$\Phi = \Phi(x, y, z, ct)$

The use of ct rather than t is dimensionally convenient because x, y, z, and ct all have dimensions of length.

The fourth coordinate is nevertheless fundamentally different from the three spatial coordinates. It represents time multiplied by the invariant speed c.

The spacetime interval is

$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2$

Equivalently,

$ds^2 = dx^2 + dy^2 + dz^2 - d(ct)^2$

The minus sign distinguishes Minkowski spacetime from ordinary Euclidean four-dimensional space.

4. MINKOWSKI METRIC

Using the metric convention

$g_{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$

the spacetime interval can be written as

$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$

The inverse metric has the same diagonal form:

$g^{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$

This metric determines how spatial and temporal derivatives combine to produce a Lorentz-invariant differential operator.

The essential feature is that the temporal component enters with the opposite sign from the spatial components.

5. THE FOUR-DIMENSIONAL WAVE OPERATOR

Define the four-dimensional derivative

$\partial_\mu = \partial/\partial x^\mu$

The Lorentz-invariant wave operator is defined by

$\Box = g^{\mu\nu}\partial_\mu\partial_\nu$

For the metric chosen above,

$\Box = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2 - \partial^2/\partial(ct)^2$

Since

$\partial/\partial(ct) = (1/c)\partial/\partial t$

we obtain

$\partial^2/\partial(ct)^2 = (1/c^2)\partial^2/\partial t^2$

Therefore,

$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2$

6. THE LORENTZ-INVARIANT CLASSICAL WAVE EQUATION

The source-free equation is

$\Box\Phi = 0$

Expanding,

$\partial^2\Phi/\partial x^2 + \partial^2\Phi/\partial y^2 + \partial^2\Phi/\partial z^2 - \partial^2\Phi/\partial(ct)^2 = 0$

Equivalently,

$\nabla^2\Phi - (1/c^2)\partial^2\Phi/\partial t^2 = 0$

Rearranging,

$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$

Thus, the conventional classical wave equation is recovered exactly.

The four-dimensional formulation therefore does not introduce a different wave equation. Instead, it reveals the spacetime structure of the equation.

7. LORENTZ INVARIANCE

A Lorentz transformation connects the coordinates of two inertial observers:

$x^\mu \rightarrow x'^\mu$

The defining property of the transformation is the invariance of the spacetime interval:

$ds'^2 = ds^2$

For a scalar field,

$\Phi'(x') = \Phi(x)$

The d'Alembertian is constructed from the Minkowski metric and therefore has invariant form.

Consequently,

$\Box\Phi = 0$

has the same mathematical form in all inertial frames.

This property distinguishes the relativistic wave equation from a generic wave equation written in an arbitrary coordinate system.

8. PLANE-WAVE SOLUTION

Consider the plane-wave solution

$\Phi = A\, e^{i(k\cdot r - \omega t)}$

where

$r = (x, y, z)$

and

$k = (k_x, k_y, k_z)$

For this field,

$\nabla^2\Phi = -k^2\Phi$

and

$\partial^2\Phi/\partial t^2 = -\omega^2\Phi$

Substitution into the wave equation gives

$-k^2\Phi + (\omega^2/c^2)\Phi = 0$

For a nonzero field,

$\omega^2/c^2 = k^2$

Therefore,

$\omega^2 = c^2k^2$

and hence

$\omega = ck$

for positive frequency.

This is the dispersion relation for the massless wave.

9. THE FOUR-WAVE-VECTOR

Define the wave four-vector by

$k^\mu = (k_x, k_y, k_z, \omega/c)$

The plane wave can be expressed in four-dimensional notation as

$\Phi = A\, e^{ik_\mu x^\mu}$

The wave equation requires

$k_\mu k^\mu = 0$

Using the adopted metric,

$k_\mu k^\mu = k^2 - \omega^2/c^2$

Therefore,

$k_\mu k^\mu = 0$

implies

$\omega^2 = c^2 k^2$

The four-wave-vector is consequently a null vector.

10. MASSLESS CHARACTER

The relativistic scalar-field equation with mass m is

$(\Box + m^2c^2/\hbar^2)\Phi = 0$

When

$m = 0$

the mass term disappears and the equation becomes

$\Box\Phi = 0$

Thus, the source-free scalar wave equation is the massless limit of the relativistic scalar-field equation.

The corresponding dispersion relation is

$\omega^2 = c^2 k^2$

The absence of a mass-dependent term is characteristic of the massless case.

11. NULL PROPAGATION AND THE LIGHT CONE

The spacetime interval is

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2$$

For a null trajectory,

$$ds^2 = 0$$

Therefore,

$$dx^2 + dy^2 + dz^2 = c^2dt^2$$

Defining

$$d\ell^2 = dx^2 + dy^2 + dz^2$$

we obtain

$$d\ell = c\,dt$$

Consequently,

$$d\ell/dt = c$$

The wave equation is therefore associated with propagation along the null structure of Minkowski spacetime.

The set of null directions forms the light cone.

12. CLASSICAL FIELD VERSUS MECHANICAL WAVE

A central conceptual issue is the distinction between a classical wave and a mechanical wave.

A mechanical wave requires a physical system whose mechanical degrees of freedom participate in the disturbance.

For example, a string displacement u may satisfy

$$\partial^2u/\partial t^2 = v^2\partial^2u/\partial x^2$$

Here u describes a mechanical displacement.

The scalar field Φ in

$$\Box\Phi = 0$$

does not necessarily have such an interpretation.

It can instead represent a fundamental classical field.

Therefore, the mathematical existence of a wave equation does not by itself imply the existence of a material medium.

13. ELECTROMAGNETIC WAVES

Electromagnetic radiation is described fundamentally by Maxwell's equations.

In vacuum, Maxwell's equations imply wave equations for the electric and magnetic fields:

□E = 0

and

□B = 0

These equations have the same d'Alembertian structure as

□Φ = 0

However, the electromagnetic field is not a scalar field. It possesses vector and tensor structure.

The electromagnetic field can be represented covariantly by the antisymmetric field tensor F_μv.

Therefore, the scalar equation should not be identified as the complete electromagnetic theory.

Rather, it demonstrates the general mathematical structure shared by massless relativistic waves.

14. LAGRANGIAN FORMULATION

The equation

□Φ = 0

can be obtained from a Lorentz-invariant classical Lagrangian.

For a free real scalar field, consider

ℒ = ½∂_μΦ ∂^μΦ

The action is

S = ∫ℒ d⁴x

Applying the Euler-Lagrange field equation produces

$\partial_\mu \partial^\mu \Phi = 0$

Therefore,

$\Box \Phi = 0$

This result demonstrates that the wave equation can be formulated consistently within classical relativistic field theory.

The field does not have to be interpreted mechanically for the variational formulation to exist.

15. ENERGY AND MOMENTUM

A classical scalar field can possess energy and momentum.

For the free scalar field, the energy-momentum tensor can be written, with the corresponding sign convention, as

$T^{\mu\nu} = \partial^\mu \Phi \, \partial^\nu \Phi - \tfrac{1}{2} g^{\mu\nu} \partial_\alpha \Phi \, \partial^\alpha \Phi$

The field equation implies the conservation relation

$\partial_\mu T^{\mu\nu} = 0$

Thus, the field can transport conserved physical quantities even though it is not interpreted as a mechanical displacement of matter.

This provides an important distinction between the concepts of field propagation and material motion.

16. SOURCE-FREE AND SOURCE-DRIVEN EQUATIONS

The present paper considers the source-free equation

$\Box \Phi = 0$

A source can be introduced through an equation of the form

$\Box \Phi = J$

Here J represents a source term, with its precise normalization depending on the physical theory and convention.

The source-free equation describes free propagation, whereas the sourced equation describes a field influenced by an external or internal source.

This distinction is analogous to the difference between free electromagnetic radiation and electromagnetic fields generated by charges and currents.

17. WHY FOUR DIMENSIONS MATTER

The expression

$\Phi = \Phi(x, y, z, ct)$

makes explicit that the field is defined over spacetime.

The wave equation then becomes

$\partial^2\Phi/\partial x^2 + \partial^2\Phi/\partial y^2 + \partial^2\Phi/\partial z^2 - \partial^2\Phi/\partial(ct)^2 = 0$

The equation cannot be understood correctly by treating all four coordinates as equivalent Euclidean directions.

The temporal coordinate has a different metric sign.

Thus, the essential structure is not simply four coordinates but rather four-dimensional Lorentzian spacetime.

This Lorentzian structure produces the causal light-cone geometry characteristic of relativistic wave propagation.

18. DISCUSSION

The four-dimensional formulation provides a deeper interpretation of the classical wave equation.

The ordinary form

$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$

emphasizes the relationship between spatial curvature and temporal acceleration.

The relativistic form

$\Box\Phi = 0$

emphasizes that the field exists on spacetime and that its differential equation is compatible with Lorentz symmetry.

The two descriptions are mathematically equivalent when the Minkowski spacetime structure is understood.

The four-dimensional form is therefore not merely a change of notation. It exposes the geometric principle underlying relativistic wave propagation.

19. LIMITATIONS

The equation

$$\Box\Phi = 0$$

should not be interpreted as a universal equation for all waves.

Different physical systems require different dynamical equations.

Mechanical waves depend on properties of matter such as density, elasticity, and pressure.

Electromagnetic waves require Maxwell's equations.

The scalar equation is therefore best interpreted specifically as the equation governing a free massless scalar field.

Its conceptual importance lies in demonstrating how wave propagation can be expressed independently of a mechanical medium.

20. CONCLUSION

The classical wave equation

$$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$$

can be naturally expressed in four-dimensional spacetime using the coordinates

$$x^\mu = (x, y, z, ct)$$

With the Minkowski metric

$$g_{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$$

the corresponding wave operator is

$$\Box = g^{\mu\nu}\partial_\mu\partial_\nu$$

and explicitly,

$$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2$$

The source-free equation is therefore

$$\Box\Phi = 0$$

A plane-wave solution,

$\Phi = A\, e^{i(k\cdot r - \omega t)}$

leads to the dispersion relation

$\omega^2 = c^2k^2$

In four-vector language, this is expressed as

$k_\mu k^\mu = 0$

showing that the wave four-vector is null.

The analysis establishes that the classical massless scalar wave equation is naturally a Lorentz-invariant equation on Minkowski spacetime. Its mathematical structure does not require a mechanical medium. Instead, it describes the propagation of a classical field whose physical interpretation must be supplied by the relevant field theory.

FINAL MATHEMATICAL SUMMARY

Spacetime coordinates:

$x^\mu = (x, y, z, ct)$

Spacetime interval:

$ds^2 = dx^2 + dy^2 + dz^2 - c^2dt^2$

Metric:

$g_{\mu\nu} = \mathrm{diag}(1, 1, 1, -1)$

Four-dimensional derivative:

$\partial_\mu = \partial/\partial x^\mu$

Wave operator:

$\Box = g^{\mu\nu}\partial_\mu\partial_\nu$

Explicit wave operator:

$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2$

Massless scalar-field equation:

$\Box\Phi = 0$

Expanded equation:

$\partial^2\Phi/\partial x^2 + \partial^2\Phi/\partial y^2 + \partial^2\Phi/\partial z^2 - \partial^2\Phi/\partial(ct)^2 = 0$

Equivalent classical form:

$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$

Plane wave:

$\Phi = A\, e^{i(k\cdot r - \omega t)}$

Dispersion relation:

$\omega^2 = c^2k^2$

Four-wave-vector:

$k^\mu = (k_x, k_y, k_z, \omega/c)$

Null condition:

$k_\mu k^\mu = 0$

Massless limit:

$m = 0 \rightarrow \Box\Phi = 0$

Central proposition:

A free massless classical scalar field propagating on Minkowski spacetime satisfies the Lorentz-invariant equation

$\Box\Phi = 0$

This preserves the original paper's structure and content while changing the scalar-field notation from φ to capital Φ.